Khoa Minh Pham

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Education

Bucknell University Expected May 2027
Bachelor of Science in Computer Science, Data Science Co-Major Lewisburg, PA
Coursework: Operating Systems Design, Computer Networks and Security, Data Structures and Algorithms

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C; object-oriented programming
Backend & Web: FastAPI, Node.js, Express, Spring Boot, RESTful APIs, React, Next.js, LangChain
Data: PostgreSQL, Redis, SQLite, ChromaDB, Spark
Systems & Tools: Concurrency, caching, WebSockets, Docker containers, Linux, Git, GitHub Actions (CI/CD), SLURM

Experience

Apache Sedona Aug 2026 – Present
Open Source Contributor *Remote*

- Contributed **4 merged pull requests** to an open-source distributed geospatial engine, resolving crashes when reading affected compressed NetCDF4 datasets from Amazon S3.
- Optimized metadata lookup performance by storing coordinate systems at object construction, eliminating distributed Spark aggregation for local Python objects; verified **zero Spark jobs** with **8 new tests**.
- Worked with maintainers through code review to align Java polygon checks with PostGIS, preserve null-input handling, and add regression tests across **4 modules**.

Bucknell University (KALMUS Research) Oct 2025 – Present
Student Software Developer *Lewisburg, PA*

- Build asynchronous film-analysis workflows with two student developers and a faculty mentor; **10+ students and researchers** have analyzed **50+ films** on a high-performance computing cluster using SLURM.
- Implemented concurrent batch uploads for multi-gigabyte films, reducing batch upload time by an estimated **30%**; reused completed results to avoid redundant computation.

JPL Integrated Communications Jun 2026 – Aug 2026
Interactive Developer Intern *Harrisburg, PA*

- Owned the Node.js/Express backend on a **three-person team**, building real-time WebSocket messaging to synchronize controllers and projectors for an exhibit now installed on site.
- Built a WordPress-to-local content pipeline with caching, scheduled refreshes, and staged updates with rollback, enabling the exhibit to run **fully offline** from cached content.

Bucknell University (Office of Institutional Research & Analytics) Oct 2025 – May 2026
Student Software Developer *Lewisburg, PA*

- Built LangChain retrieval over ChromaDB and SQLite-backed conversation memory on a **three-person team**, grounding cited chatbot answers in a **491-page catalog** and **17 policy documents**.
- Implemented concurrent AI model calls and document searches with Python asyncio in FastAPI backend services; streamed partial responses through server-sent events so users could read before generation finished.

Parker Dewey Feb 2026 – Apr 2026
Software Engineering Intern (Client Engagement) *Lewisburg, PA*

- Delivered a cybersecurity assessment platform as **sole developer**, working with the founder and a faculty mentor to build RESTful APIs, authentication, Stripe billing, and asynchronous AI reports.
- Implemented tenant-scoped data authorization with PostgreSQL queries and AES-256-GCM field encryption; mapped **5 security domains** to **0–100 readiness scores** and **4 risk tiers**.

Projects

Freshness Tracker | SASEHack 2026 *React Native, Expo, FastAPI, ThingSpeak, Ollama*

- Won **Best First Hack** on a **4-person team**; built inventory, label-scanning, and assistant screens and integrated teammates' hardware and freshness engine into a **working demo**.
- Connected **minute-by-minute** fridge sensor readings to FastAPI backend services and added tests, updating freshness estimates with measured temperatures.

SpotOn | Music Research App *React, TypeScript, Groq, Redis, Vercel*

- Deployed a music research app on Vercel, combining Spotify, Genius, and web sources into AI-generated reports with citations.
- Cached reports in Redis to avoid repeated requests and AI processing; streamed live progress during research.